# UniTrade - NFR Evidence Pack

Project: UniTrade | Commit: 9340c2936bfa | Generated (UTC): 2026-10-22 12:12:52 UTC
21 requirements across 7 SRS quality attributes | pass: 19, documented: 1, partial: 1
Validation: PASS - all evidence references resolve
Evidence labels are clickable (committed files link to the commit permalink; CI items link to the test file and the workflow).

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## Administration

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-07a | Moderation queue returns cases oldest-first | oldest-first ordering | Integration test | pass | test: AdminSpineTests.cs<br>CI: Backend CI |
| QR-07b | Every admin decision is audited | actor + timestamp + reason present (exactly one row) | Integration test | pass | test: AdminSpineTests.cs<br>CI: Backend CI |
| QR-07c | Decision notifies the subject | Notification dispatched on each decision | Integration test (notification check) | pass | test: AdminDecisionTests.cs<br>CI: Backend CI |
| QR-07d | SLA breach is surfaced on the dashboard | Cases older than SLA (48 h verification / 72 h disputes) flagged | Integration test | pass | test: AdminDecisionTests.cs<br>CI: Backend CI |

## Maintainability

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-05a | Critical-path tests + static-analysis gate on main | Critical business paths covered by unit + integration tests; static-analysis gate runs in CI | SonarCloud (static analysis) + xUnit (critical-path suites) | partial | Backend CI / Web Frontend CI |
| QR-05b | Migrations apply cleanly from an empty database | Applies clean on a fresh Postgres | dotnet ef database update | pass | CI: Backend CI |

## Performance

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-01a | p95 read latency (feed / search / browse) under load | p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline) | k6 | pass | perf-summary(QR-01a).json |
| QR-01b | Chat message delivery latency | p95 <= 500 ms | Playwright (SignalR round-trip timing) | pass | chat-latency.json |

## Reliability

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-04a | Duplicate payment notification is idempotent | Exactly one side effect (one transaction, status completed) | Integration + duplicate delivery | pass | test: PaymentWebhookTests.cs<br>CI: Backend CI |
| QR-04b | A single failed worker cycle does not stop the worker | Worker continues to next cycle after an injected fault | Integration + fault injection | pass | test: ReservationExpiryWorkerTests.cs<br>CI: Backend CI |
| QR-04c | A failed deployment auto-rolls-back | Health-gated: health check before traffic shift; automatic rollback on failure | Deploy config + observed failing run | documented | cd.yml<br>QR-04c_ Failed deployment should automatically roll back to the last live revision.pdf |
| QR-04d | Availability over the 30 days before the demo | >= 99.5% | UptimeRobot | pass | UpTimeRobotEvidence.png |

## Scalability

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-02 | Error rate + latency at 2x baseline load | p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs) | k6 | pass | scale-summary(QR-02).json |

## Security

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-03a | Admin endpoints reject non-admin callers | 100% reject (anon -> 401, student -> 403, admin -> 200) | Integration test | pass | test: AdminSpineTests.cs<br>CI: Backend CI |
| QR-03b | Forged/tampered payment callback has no effect | Rejected (400 invalid_signature), no state change | Integration + fault injection | pass | test: PaymentWebhookTests.cs<br>CI: Backend CI |
| QR-03c | OTP storage + comparison | Peppered hash; constant-time compare; never equals raw OTP | Unit test | pass | test: OtpSecurityTests.cs<br>CI: Backend CI |
| QR-03d | No medium-or-above exposure on staging (ZAP baseline) | 0 high/critical; mediums triaged | OWASP ZAP baseline | pass | zap-report(backend-QR-03d).html<br>zap-report(backend-QR-03d).json<br>zap-report(frontend-QR-03d).html<br>zap-report(frontend-QR-03d).json |
| QR-03e | No high/critical dependency CVEs on main | 0 high/critical | dotnet list package --vulnerable + npm audit --audit-level=high | pass | npm-audit-report.txt<br>dotnet-vulnerability-report.txt |
| QR-03f | No high/critical OS or library CVEs in container images | 0 high/critical (triaged entries listed in .trivyignore) | Trivy (container image scan) | pass | trivy-backend.txt<br>trivy-frontend.txt |

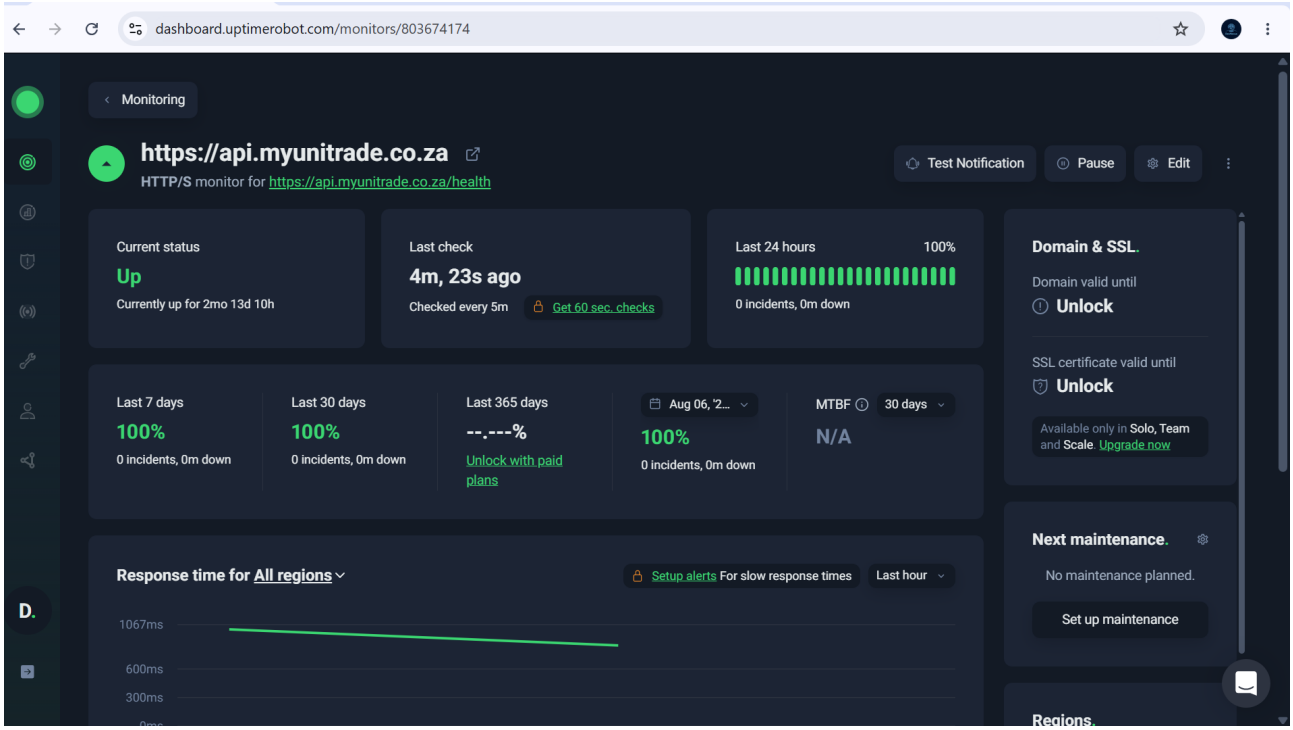
## Usability

| ID | Requirement | Target | Tool | Status | Evidence |
|---|---|---|---|---|---|
| QR-06a | Accessibility on public + admin pages | 0 serious/critical (Lighthouse/axe >= 90 threshold) | Lighthouse (axe-core) | pass | perf-access-lighthouse/ (9 files) |
| QR-06b | Core task completes end-to-end in <= 2 min | <= 120s {browse -> reserve -> chat -> meetup -> initiate payment} | Playwright e2e | pass | core-task.json |

## Key measurements

| ID | Metric | Measured | Target |
|---|---|---|---|
| QR-01a | p95 read latency | **43.8 ms** | p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline) |
| QR-01a | request error rate | **0.00%** | p95 <= 800 ms, error rate < 1% at ~40 concurrent users (baseline) |
| QR-02 | baseline p95 | **51.4 ms** | p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs) |
| QR-02 | peak p95 (2x load) | **65.0 ms** | p95(2x) <= 1.5x baseline p95, error < 1% (baseline 40 VUs -> peak 80 VUs) |

## Availability evidence (QR-04d)




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Integrity hashes for every committed evidence file are in tests/nfr/evidence/SHA256SUMS (verify with `sha256sum -c`).